

Hands-On: Understanding Props, Default Props, and Prop Types in React

Requirements

- Basic knowledge of React
- Text editor and web browser
- Node.js and npm installed on your computer

Instructions

1. Create a new React project using **create-react-app**.

```
npx create-react-app props-lab  
cd props-lab
```

2. Open the project folder in your text editor.
3. In the `src` folder, create a new file named `PropsComponent.js`.

```
import React from "react";  
  
function PropsComponent(props) {  
  return (  
    <div>  
      <h1>Hello, {props.name}!</h1>  
      <p>Your age is {props.age}</p>  
      <p>Your country is {props.country}</p>  
    </div>  
  );  
}
```

```
export default PropsComponent;
```

4. In the `App.js` file, import the `PropsComponent` and render it in the `render()` method.

```
import React from "react";  
import PropsComponent from "./PropsComponent";  
  
function App() {  
  return (  
    <div className="App">  
      <PropsComponent name="John" age={30} country="USA" />  
      <PropsComponent name="Alice" age={25} country="Canada" />  
    </div>  
  );  
}  
  
export default App;
```

5. Start the development server.

`npm start`

6. Open your browser and go to `http://localhost:3000/`.
7. You should see the following output:

Hello, John!
Your age is 30.
Your country is USA.

Hello, Alice!
Your age is 25.
Your country is Canada.

This means that the `PropsComponent` is receiving props from the `App` component and rendering them.

8. Now, let's add default props to the `PropsComponent` in case some props are not passed. Add the following code at the end of the `PropsComponent` file:

```
PropsComponent.defaultProps = {  
  age: 18,  
  country: "Unknown",  
};
```

This means that if the `age` and `country` props are not passed, their default values will be 18 and "Unknown", respectively.

9. Save the file and refresh the browser. You should see the following output:

Hello, John!
Your age is 30.
Your country is USA.

Hello, Alice!
Your age is 25.
Your country is Canada.

Hello, !
Your age is 18.
Your country is Unknown.

Hello, !
Your age is 18.
Your country is Unknown.

This means that the default props are used for the last two `PropsComponent` instances where `name` is not passed.

10. Finally, let's add prop types to the `PropsComponent` to ensure that the correct data type is passed for each prop. Install the `prop-types` package:

`npm install prop-types`

11. In the `PropsComponent` file, import the **PropTypes** from the `prop-types` package and add the following code at the end of the file:

```
import PropTypes from "prop-types";

PropsComponent.propTypes = {
  name: PropTypes.string.isRequired,
  age: PropTypes.number,
  country: PropTypes.string,
};
```

This means that the `name` prop is required and must be a string, and the `age` and `country` props are optional but must be numbers and strings, respectively.

12. Save the file and refresh the browser. You should see the same output as in step 9, but with the addition of prop types, React will now show warnings in the console if the wrong data type is passed for any prop. For example, if you change the `age` prop to a string in the `App.js` file like this:

```
<PropsComponent name="John" age="30" country="USA" />
```

You should see the following warning in the console:

```
Warning: Failed prop type: Invalid prop `age` of type `string` supplied to `PropsComponent`, expected `number`.
```

This means that the `age` prop should be a number, not a string.

That's it! You've successfully completed the hands-on lab for defining and passing props, accessing props in components, default props, and prop types in React. You can now use these concepts to build more complex React applications with reusable components that can accept different sets of props.